

# Chloe Kim

## Resume – Academic Emphasis

📍 Cranbrook Kingswood Upper School, 39221 Woodward Ave. Bloomfield Hills, MI

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### Academic Interests & Selected Outputs

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*Focus:* AI × Human Decision-making under Uncertainty & Fairness

- Cognition & intelligence — perception, reasoning, risk
- Applied domains: social applications of AI, AI insurance, legal networks, healthcare triage

*Keywords:* Computational cognitive science; reinforcement learning & Bayesian inference; NLP; causal inference

*Selected Outputs*

- *Professional Academic Presentations:* EURO 2025; Helmholtz AI 2025 (*Spotlight*)  
— Human *vs.* model-based reasoning; recognition & uncertainty, fairness; AI applications
- *Patent:* AI insurance framework linking reliability risks to actuarial metrics (Sole Inventor)
- *Software Reg. (4 works):* Network dynamics, power-law & inequality study (Sole Developer)

### Education

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**Cranbrook Kingswood Upper School** *High school student*

*Aug 2022–  
Bloomfield Hills, MI*

Humanities, Biology, Linguistics, AI, Comparison of learning algorithms

**Harvard University** *Undergraduate-level studies*

*July 2024–Aug 2024  
Cambridge, MA*

Molecular Biology (BIOS-S1a) (A-) and Historical Linguistics (LING-120) (A)  
Studies at Harvard Summer School (for credit)

南京时代双语学校 (Nanjing Times College) *Middle school student*

*Apr 2020–Jun 2022  
南京(Nanjing), China*

**Kyunghee School** *Elementary school student*

*Mar 2014–Feb 2020  
Seoul, Republic of Korea*

### Selected Research & Intellectual Property

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**Patent** *Sole Inventor*

“System and method for evaluating AI output consistency and providing dynamic insurance”  
#10-2025-0104581, patent pending (Korea)

- Quantifies AI risks (factuality, logical/semantic consistency, bias) into actuarial metrics.
- Designs dynamic premium pricing & claim validation linked to AI performance.
- *AI Insurance* – recompensation for AI hallucination

**Software Registrations** *Sole Developer*

“Interpreting Social Intent and Behavior: Temporal Dynamics of Network Heatmaps” (#035490)

- Analyzes temporal networks, creating centrality- & time-based maps of social relations.  
“Analysis of Social Intentionality: Visualization of Case-law Networks” (#035307)
- Processes legal datasets, computing centrality & visualizing case-law structures.  
“Simulation of Behavioral Relationships in Power-law Dynamics using AI” (#035489)
- Models behavior relations under power-law dynamics with AI sampling & visualization.  
“Inequalities in Human Behavior: A Graph-based and AI Interpretation” (#034662)
- Analyzes behavioral inequality using AI/graphs, quantifying disparities via references.

### Research Presented *First author & Presenter*

“Dynamic Network Analysis of Legal Citation Networks (especially medical)”

*34th European Conference on Operational Research (EURO 2025), Leeds, UK*

- Analyzed 86,000+ Supreme Court cases using network and power-law analysis, showing concentrated reliance on landmark precedents.
- Illustrate how structural citation patterns shape legal & medical decision-making.

“Emergent Architecture of Law: Dynamic Network and AI Analysis of Comprehensive and Medical Legal Precedents in South Korea (1951–2023)”

*Submitted to PLOS One (w/ S. Kim), Oct 2025*

“An Experimental Analysis of Intuition and Semantic Reasoning in Linguistic Visualizations by Humans and AI”

*Helmholtz AI Conference 2025, Karlsruhe, Germany*

- Compared 40+ multilingual participants with GPT-4 on sound-to-shape tasks, revealing humans’ intuitive mappings *vs.* AI’s semantic reasoning.
- Findings support dual-mode cognition, highlighting differences in human and machine decision-making.

“Multimodal Deep Learnings and Interpretability for Automated Mallampati Scoring and Endoscopy Prediction” (*Spotlight*)

*Helmholtz AI Conference 2025, Karlsruhe, Germany*

- Trained on 500+ patient cases with fused imaging (Vision Transformer) and clinical features (XGBoost), achieving 92.7% accuracy and AUROC 0.90.
- Results demonstrate scalable, interpretable AI for airway management, highlighting contrasts between human judgment and standardized decision-making.

## Work Experience

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**Lawfirm Hoam** *Research Assistant with Lawyer*

*Summer 2024–  
Seoul, Rep. of Korea & Remote*

- Perform data analytics and lawsuit cases research with AI and quant. network analyses.
- Presented results in an academic conference.

**Quilt.ai** *Research Intern*

*Jul 2024–Dec 2024  
Boston, MA & Remote*

- Evaluated and researched AI narrative generation for fashion ads. that support human perception and cognition
- Ran market surveys; produced insight memos used by the design/product teams.

**Media art Yeseung Lee's team** *Assistant Editor/Supporter**Dec 2024–  
Seoul, Rep. of Korea & Remote*

- Supported a project on art and human cognition.
- Edited artist statements and exhibition texts (Singapore, Seoul Dongdaemun Design Plaza (DDP), National Museum of Modern Art (MMCA) exhibitions)

**Elroi AI** *Research Intern**May 2025~  
Troy, MI & Remote*

- Prototyped and studied modules for generation and HAI workflows; conducted user tests
- Delivered a technical brief.

**SAT Mathematics Tutoring** *Tutor**June 2025~September 2025  
NYC, NY & Remote*

- Tutored high school students in SAT Math preparation and strategies

**Computer Science Projects** *Independent researcher**Jun 2022–  
Bloomfields Hills, MI & remote*

- Built coding projects including BMI Calculator, Quiz Game, Server–Client Trading Apps, and AI&ML-based game simulations
- 9th – 12th grade

**Leadership & Service**

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**Varsity Rowing Team** *Coxswain*

- Coach/manage boats (steering/making calls & corrections), mentor novice rowers, delegate roles, manage race day logistics & encourage team support

**Linguistics Club** *Founder, Co-President*

- Founded Linguistics Club to promote different branches of linguistics; raise awareness of diverse languages; NACLO prep; AI+ Linguistics research

**Ethics Bowl Club** *President/Founder*

- Founded club; recruited, hosted state-wide practice events, organized spaces for ethical thinking & dialogue in & out of school & community

**Crane Clarion (School newspaper)** *Assoc. Arts Editor, Graphic Designer, Staff Writer*

- Led artists & writers; mentored junior writers; established journalistic protocols; publicized arts & school spirit; wrote cultural sens. guidelines

**LITE (Leaders in the Environment)** *Food Board Captain*

- Liaison to school meal providers; promoted inclusive meals; systemized negotiation process btw. student body, faculty, and providers; org 10+ events

**Cranbrook Kingswood Chamber Orchestra, Thornlea Quartet** *Violinist; Organizer*

- Violinist in school orchestra; performed at school/community; organized fundraising concerts for Brilliant Detroit; mentored peers

**Tutoring** *Comm. Service – Volunteer tutor*

- Independent & Volunteer – Local & Overseas; hybrid—in-person/online via Zoom/Teams)

- Tutored 5-7 under-resourced students: English (Chinese & Korean) + SAT Math (US); ran Zoom lessons; built practice sets; measurable score gains

**Gold Key** *Sophomore Lead*

**“Ergasterion” – Theater & Drama club** *Stage Crew & Member*

**Student Council** *Member*

**World Scholars Cup** *Team leader*

- Led the debate team; won in conferences

## Awards and Honors

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**National Latin Exam Cum Laude and 2x Magna Cum Laude** – *G9, 10, 11; National*

**Ethics Bowl Central Division 4th place** – *G10, 11; State*

**3rd in State for National Economics Challenge** – *G10; State*

**District MSBOA 1st Division (solo) & 1st/2nd Division (quartet)** – *G9, 10, 11; State*

**NEC – National Economics Challenge; 3rd (D. Ricardo)** – *G10; State*

**Strickland Writing Award** – *G9; School*

**Dean’s List & Honors** – *G9, 10, 11; School*

**AP Scholar with Distinction Award** – *G11; School*

**World Scholars Cup: Qualified for the global round** – *2020-21; National (China)*

## External Studies

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**Human Asia Academy**

*Jun 2023,  
Seoul, Rep. of Korea*

- “Human rights” for North Korean people

**MIT OpenCourseWare** – *Self Studies*

- Python & Julia programming language; concepts, grammar, applications
- Abstraction, object-oriented programming, recursion, debugging, interface to C

**FastCampus Korea** – *Machine Learning Courses*

- AI, Machine learning and Data Analytics (Python, Tensorflow, Dockers, LLM-agents)

## Skills

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**AI and Data Analytics**

- Statistics & Data analytics
- Network and complex system analyses

**Technical Computer Science** *Java, Python, Julia*

- Applications to Semantics & philosophy

## Additional

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Citizenship: United States

Languages: English (native); Korean (native/bilingual); Chinese (Mandarin) (native); French (intermediate); Japanese (intermediate); Latin (NLE Latin III)